Three-Root Algebraic Components at Distance One Half

Candidate counterexample for arXiv:1601.01505
Run `fa_banach_001`, lane 0

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Status. Candidate counterexample, likely valid, pending specialist human review. For $p(\lambda) = \lambda(\lambda-1)(\lambda-2)$ in $M_2(\mathbb{C})$, two distinct components of $E_p(M_2(\mathbb{C}))$ have distance exactly $1/2$, while the source conjectures the lower bound 1.

1 Source conjecture

For a unital complex Banach algebra A and a polynomial

$$p(\lambda) = \prod_{j=1}^{n} (\lambda - \lambda_j)$$

with distinct roots, Makai and Zemánek write

$$E_p(A) = \{a \in A : p(a) = 0\}.$$

They conjecture that any two distinct connected components C_1, C_2 of $E_p(A)$ satisfy

$$\operatorname{dist}(C_1, C_2) \geq \min_{i \neq j} |\lambda_i - \lambda_j|. \tag{1}$$

They state the analogous conjecture for self-adjoint algebraic elements in a C^*-algebra [1, p. 5].

2 Proof intuition

With three roots, two components can have spectral multiplicities whose traces differ by only one even though their occupied nonzero roots differ. We choose rank-one idempotents e, f so that the difference between the root-1 element e and the root-2 element $2f$ has the form

$$2f - e = \frac{1}{2}I + N,$$

where N is nonzero but square-zero. A nonunitary similarity can shrink a 2×2 nilpotent Jordan block by an arbitrary factor. Simultaneously conjugating e and $2f$ preserves their connected components, while their difference tends to $I/2$. The trace difference supplies the matching lower bound $1/2$.

Theorem 9. *There exists an explicit constant $c(\lambda_1, \ldots, \lambda_n) > 0$ (depending on $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, and invariant under any map $(\lambda_1, \ldots, \lambda_n) \mapsto (a+b\lambda_1, \ldots, a+b\lambda_n)$ with $a, b \in \mathbb{R}$ and $b \neq 0$) such that the following holds. If A is a unital complex C^*-algebra, and C_1, C_2 are distinct connected components of $S_p(A)$, then the distance of C_1 and C_2 is at least $c(\lambda_1, \ldots, \lambda_n) \cdot \min\{|\lambda_i - \lambda_j| \mid 1 \leq i, j \leq n,\ i \neq j\}$.*

Conjecture. Let A be a unital complex Banach algebra (C^*-algebra) and C_1, C_2 distinct connected components of $E_p(A)$ (of $S_p(A)$). Then the distance of C_1 and C_2 is at least $\min\{|\lambda_i - \lambda_j| \mid 1 \leq i, j \leq n,\ i \neq j\}$.

For $n = 2$ this conjecture is equivalent to the statement that this distance for $E_p(A) := E(A)$ (for $S_p(A) := S(A)$) is at least 1, which is due to J. Zemánek [Ze] (due to S. Maeda [Mae]). For $n \geq 3$ we do not even know whether this distance for the Banach algebra case is positive.

Figure 1: Theorem 9 and the conjecture in arXiv:1601.01505, PDF page 5.

3 Counterexample

Equip $A = M_2(\mathbb{C})$ with its usual operator norm and take

$$p(\lambda) = \lambda(\lambda - 1)(\lambda - 2).$$

Let C_{01} denote the component of elements with root multiplicities $(1, 1, 0)$ and let C_{02} denote the component with multiplicities $(1, 0, 1)$, in the root order $(0, 1, 2)$.

Theorem 1. *The components C_{01} and C_{02} are distinct and*

$$\operatorname{dist}(C_{01}, C_{02}) = \frac{1}{2}.$$

Since the minimum separation of the roots $0, 1, 2$ *is* 1*, this disproves* (1) *for the Banach-algebra set* $E_p(A)$*.*

Proof. Set

$$e = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} \frac{9}{8} & 1 \\ -\frac{9}{64} & -\frac{1}{8} \end{pmatrix}.$$

A direct multiplication gives $e^2 = e$ and $f^2 = f$, and both idempotents have rank one. Thus $e \in E_p(A)$ has root multiplicities $(1, 1, 0)$, whereas $2f \in E_p(A)$ has root multiplicities $(1, 0, 1)$.

For completeness, these multiplicities are constant on connected components. Indeed, for $a \in E_p(A)$ the Lagrange spectral idempotents

$$q_j(a) = \prod_{k \neq j} \frac{a - \lambda_k I}{\lambda_j - \lambda_k} \qquad (j = 0, 1, 2) \tag{2}$$

depend polynomially on a. Their ranks are locally constant, hence constant on every connected set. Since the rank triples for e and $2f$ differ, they belong to distinct components.

Now calculate

$$D := 2f - e = \begin{pmatrix} \frac{5}{4} & 2 \\ -\frac{9}{32} & -\frac{1}{4} \end{pmatrix} = \frac{1}{2}I + N,$$
$$N = \begin{pmatrix} \frac{3}{4} & 2 \\ -\frac{9}{32} & -\frac{3}{4} \end{pmatrix}.$$

Then $N \neq 0$ and $N^2 = 0$. More explicitly, with

$$P = \begin{pmatrix} 8 & 0 \\ -3 & 4 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

we have $P^{-1}NP = J$.

For $t > 0$, define

$$S_t = P \begin{pmatrix} t & 0 \\ 0 & 1 \end{pmatrix} P^{-1} = \begin{pmatrix} t & 0 \\ \frac{3(1-t)}{8} & 1 \end{pmatrix}.$$

Then S_t is invertible and

$$S_t N S_t^{-1} = tN. \tag{3}$$

Put

$$a_t = S_t e S_t^{-1}, \qquad b_t = 2S_t f S_t^{-1}.$$

The group $GL_2(\mathbb{C})$ is path connected, so similarity by S_t connects a_t to e inside $E_p(A)$ and connects b_t to $2f$ inside $E_p(A)$. Therefore $a_t \in C_{01}$ and $b_t \in C_{02}$. Equations above give

$$b_t - a_t = S_t D S_t^{-1} = \frac{1}{2}I + tN,$$

and hence

$$\operatorname{dist}(C_{01}, C_{02}) leq \lim_{t \downarrow 0} \|b_t - a_t\| = \frac{1}{2}.$$

For the reverse inequality, every $a \in C_{01}$ has ordinary matrix trace $\operatorname{Tr}(a) = 1$, while every $b \in C_{02}$ has $\operatorname{Tr}(b) = 2$. Since

$$|\operatorname{Tr}(x)| \leq 2\|x\| \qquad (x \in M_2(\mathbb{C})),$$

we obtain

$$\|b - a\| \geq \frac{|\operatorname{Tr}(b-a)|}{2} = \frac{1}{2} \qquad (a \in C_{01},\ b \in C_{02}).$$

The two bounds prove the asserted exact distance. □

Remark 1 (A fixed pair already violates the conjecture)**.** *The square-zero matrix N has operator norm $73/32$. Taking $t = 1/10$ in the construction gives*

$$\|b_{1/10} - a_{1/10}\| \leq \frac{1}{2} + \frac{1}{10}\frac{73}{32} = \frac{233}{320} < 1.$$

Thus one need not pass to an infimum to exhibit a cross-component pair closer than the minimum root separation.

4 Verification notes

1. **Exact algebraic identities.** The idempotency of e, f, the identity $N^2 = 0$, and the conjugation formula (3) are rational 2×2 matrix calculations.

2. **Component invariant.** Formula (2) avoids any appeal to a delicate global classification: the ranks of the spectral idempotents alone prove that the components are distinct.

3. **Component membership of the families.** The paths come from paths in $GL_2(\mathbb{C})$ and therefore remain in $E_p(A)$ under similarity.

4. **Sharp lower bound.** The trace estimate applies to every pair from the two components, not only to the constructed families.

5. **Supplementary code.** The script `code/verify_matrices.py` checks the rational identities symbolically and samples the norm convergence. It is not a dependency of the proof.

5 Scope and limitations

The counterexample disproves the Banach-algebra $E_p(A)$ assertion, even with the finite-dimensional C^*-algebra $A = M_2(\mathbb{C})$ as ambient algebra. The elements a_t, b_t are generally not self-adjoint because S_t is not unitary. Consequently, this packet does *not* disprove the separate self-adjoint conjecture for $S_p(A)$. It also does not decide the weaker question, mentioned by the source authors, whether distinct Banach-algebra components always have some positive separation depending on the roots.

6 Novelty check

A bounded search on 9 August 2026 covered:

- the run's lightweight indexes and local 2000–2026 arXiv source corpus;
- exact and close-variant web/arXiv searches using the source title, authors, "distance of connected components", and the $E_p(A)$ notation;
- the authors' expanded 2018 preprint, which repeats the same distance problem [2];
- the OpenAlex citation graph of DOI `10.1007/s10587-016-0294-6`. It contained two substantive citing works, on idempotents and similarity-orbit commutativity; neither search result addressed the component-distance conjecture.

No prior explicit counterexample or resolution was found. This bounded search does not prove novelty, and the candidate should be checked by a specialist.

7 Human-review recommendation

Recommendation: high-priority verification. The construction is finite-dimensional and the proof is short. A reviewer should verify the four matrix identities, confirm that the spectral-idempotent rank triples separate the components, and check that simultaneous similarity produces paths within the two fixed components.

References

[1] E. Makai, Jr. and J. Zemánek, *Nice connecting paths in connected components of sets of algebraic elements in a Banach algebra*, Czechoslovak Math. J. **66** (2016), 821–828; arXiv:1601.01505.

[2] E. Makai, Jr. and J. Zemánek, *On the structure of the set of algebraic elements in a Banach algebra and their liftings*, arXiv:1807.01552 (2018).